



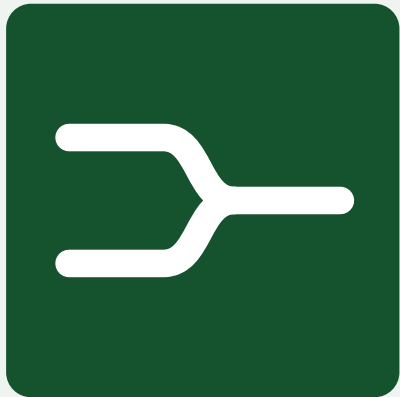
ReconMed

Automated Medicine Reconciliation

Reconciled before the visit.

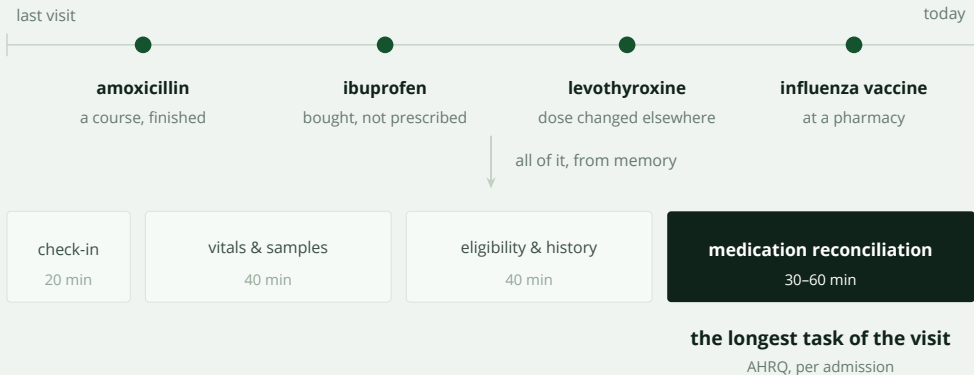
Pre-visit medication reconciliation
for clinical trial sites.

HackMIT 2026



The medication list is stale, and the visit is now

Every medication a participant takes besides the study drug is collected once, at the visit.



Medication reconciliation happens at the visit, so the record is whatever the participant remembers.

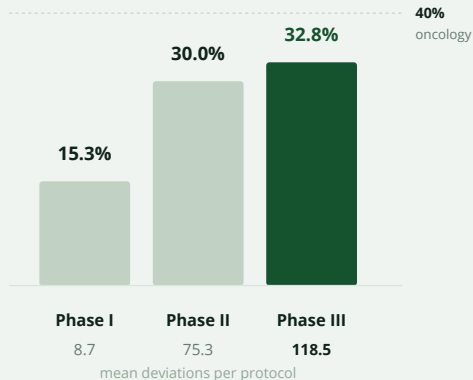
Nothing above reaches the site any other way. A course that ended six weeks ago is the easiest thing in the world to forget.

Reconciliation times: AHRQ *Patient Safety and Quality*; time-and-motion study of 103 reconciliations, *BMC Health Serv Res* 13:485. Hospital settings.

Common in every trial, and expensive once it is cold

Participants with a protocol deviation

187 protocols, 20 sponsors and CROs — Tufts CSDD, 2022



Found later, in retrospective review

One phase 2 trial — I-SPY COVID, 2025

\$4,862

per data error identified

\$6.1M

and 61,073 person-hours, to check 30% of one trial

0.36%

of the fields checked actually held an error

Calling participants between visits already works

25,829 pharmacist calls across 12 clinical trials — Martinez et al., 2017

45.5%

of those calls produced at least one intervention

Around 55% were rated high-impact — affecting trial outcomes or adherence.

Participants with an issue found by phone

adverse events



medication use against protocol



concurrent medications



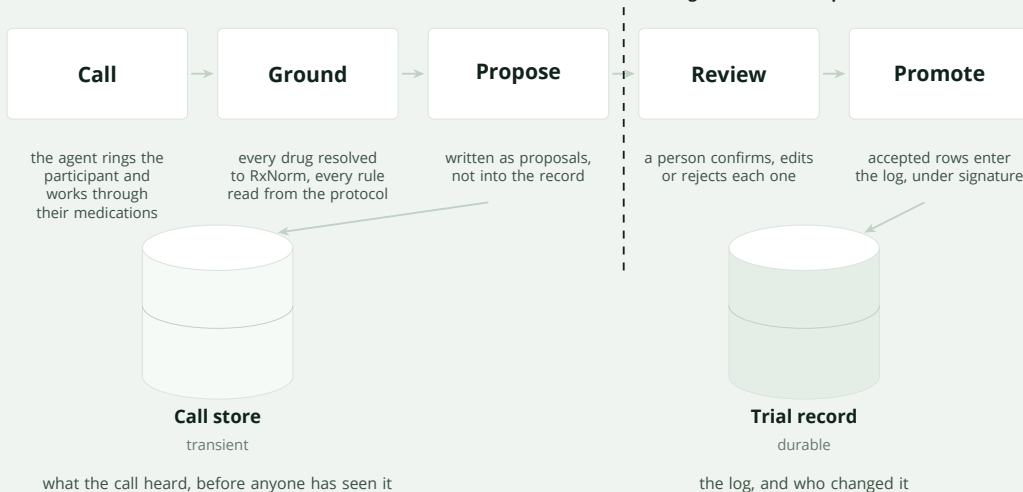
A pharmacist made every one of those calls.

That is why it is not standard practice. ReconMed makes the call, and a coordinator reviews only what it found.

Martinez et al., *Clinical Therapeutics* 39(3), 2017. 3,873 participants; 12 trials in type 2 diabetes.

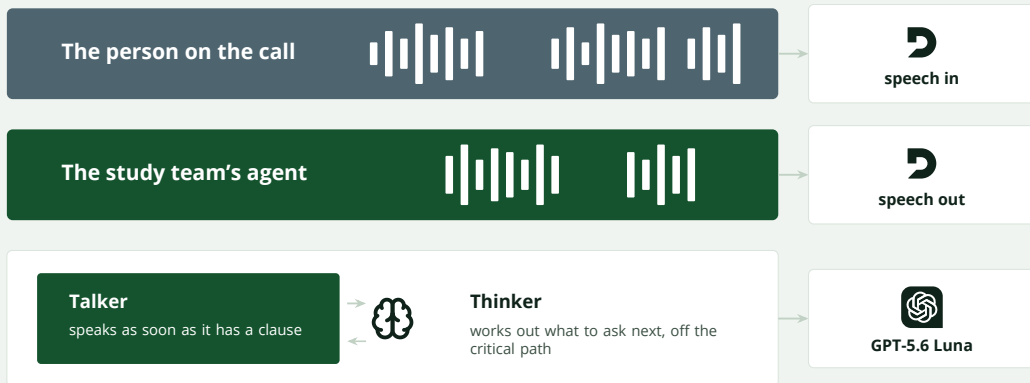
How a call becomes a record

The agent proposes. A coordinator decides.



Inside the voice agent

One model speaks. A second thinks between turns, so the line is never waiting.



It may not answer from memory. Three questions, three sources:

Is this drug banned here?

the trial's own protocol

What side effects are known?

the drug's FDA label

Is alcohol or smoking allowed?

the protocol's non-drug rules

What we do about hallucination

The model is never the source. It may not state a fact it did not look up.



GPT-5.6 Luna

decides what to ask,
never what is true



RxNorm · RxClass

what a drug is, and which class it belongs to



the drug's FDA label

which side effects are actually documented



the participant's record

the only source of anything about this person

heard

"prednasone"

resolved · RxCUI 8640

prednisone

classified

ATC H02

rule

\$5.7.2

Review queue

6 proposed changes



Prohibited

Prednisone
20 mg daily · oral

Protocol §5.7.2



matched drug
prednisone

matched class
systemic corticosteroid
ATC H02

protocol section
§5.7.2
excluded above 10 mg/day

What is on file, and what the call proposed

Review queue

7 proposed changes · 2 prohibited

on file

from the call

Add

—



Atorvastatin 20 mg nightly



pending

Modify

Metformin 500 mg twice daily



Metformin 1000 mg twice daily



accepted

Stop

Ibuprofen 400 mg as needed



stopped — GI bleed reported



rejected

each row: confirm, edit, reject · and “why” opens the tool trace

Promote 1 accepted

Nothing has to leave the machine

Three real published protocols, 25 participants — trials real, participant data synthetic.

The record



conmed.db

the trial record



patient.db

the call in progress



drugs.db

RxNorm · RxClass



general_health.db

plain-language guidance

The voice, too

speech in

Parakeet 0.6B

NVIDIA · NeMo

speech out

Qwen3-TTS 0.6B

open weights

Runs on CPU. No API key.

A participant's voice never reaches a vendor.

Or a server

```
DATABASE_URL=  
postgres://...
```

one environment
variable, same code

MIT · open

The demo runs the hosted speech path. The local models are built and round-trip tested in `local-ai/`; switching is configuration, not a rewrite.



ReconMed

Automated Medicine Reconciliation

Thank you · questions

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